

Brian S. McKinley

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EDUCATION

University of Central Florida

Orlando, FL

B.S. Electrical Engineering with Honors, GPA: 3.2

Expected Fall 2027

- Electrical Coursework: Electronics I/II; Linear Circuits I/II; Electromagnetic Fields; Embedded, Digital and Real-Time Systems; Linear Control Systems; Engineering Statistics

Lake-Sumter State College

Leesburg, FL

Associate of Arts in General Studies with Honors, GPA: 3.7

May 2023

EXPERIENCE

Undergraduate Cleanroom Fellow / Co-op

Orlando, FL

UCF Cleanroom Fellowship (Intel Scholar)

Jan 2026 - Present

- Operated AJA RF/DC PVD sputtering systems, e-beam evaporator, cryo-sputtering system, and ANRIC ALD system for thin-film deposition; performed chamber cleaning and maintenance
- Performed photolithography on 2-4 in wafers: spin coating, baking, development, edge isolation, and patterning on Heidelberg DWL 66+ laser system; designed masks in KLayout
- Characterized 5+ thin-film oxide diodes via I-V sweeps (-3V to +8V) on Keithley 2450 SourceMeter to isolate device variation, supporting graduate research on transparent semiconductors
- Designed and built an ESP32-based cleanroom chiller monitoring system (flow sensor, temperature probe, standalone Wi-Fi access point) with custom sensor plumbing, from concept through deployment

Undergraduate Learning Assistant, ECE Design

Orlando, FL

University of Central Florida, Dept. of ECE

Feb 2026 - Present

- Root-caused and recovered 30+ microcontroller and development boards (Raspberry Pi 4, MSP430, ESP-based) by rerouting 11-20 mil traces, adding bypass wiring, and clearing solder shorts
- Taught PCB layout in Fusion 360 from 13 mm to multi-foot boards; coached students on power regulators, sensor integration, ADC setup, and embedded debugging in CCS

Instrumentation and Test Engineer

Orlando, FL

Material Properties Characterization Lab, UCF

2025 - 2026

- Designed custom DB15 adapters using high-impedance TI INA122P amplifiers for analog extensometry signal acquisition in material strain testing and authored technical documentation

PROJECTS

Ancillary Building Electrical System Installation: Engineered and installed a 120/240 VAC split-phase system powering 900 sq. ft. of building infrastructure; passed NEC inspection on first pass

Ultrasonic Distance Meter: Designed an MSP430-based PCB in Fusion 360 with an ultrasonic sensor, power regulation, and LCD output.

TECHNICAL SKILLS

Semiconductor Processing: Device manufacturing and test, process engineering, design of experiment

Assembly & Packaging: SMT/THT soldering, pick-and-place, reflow, PCB fabrication, PCB schematic design, electroplating

Metrology & Test: Oscilloscopes, waveform generators, spectrum analyzers, extensometry, bench testing, circuit design and analysis

EDA & CAD: KiCad, Fusion 360, SolidWorks (CSWA), Xilinx Vivado, LTspice, Siemens Expedition, Code Composer Studio

Programming & Languages: C, Python, Embedded C, MATLAB, MIPS,

Methods & Compliance: SOP authoring, NEC compliance